BL-29 / BL-31 - Defects and retests

Date: 29 October 2026. Tracking: defect #48, linked to #29 and #31. These findings were corrected in the verification branch. Report each final execution outcome in the execution record.

Finding / trigger	Correction	Retest evidence
Present/Absent report filtering removed marks from the denominator, allowing a present-only group to report 100%	Select groups matching the status while aggregate totals still include every scoped mark; build HTML/PDF totals from the same grouped data	`AttendanceAlertAndReportIntegrationTests.test_status_and_student_report_filters_preserve_the_real_denominator` passed
A report request could contain a foreign assignment or student ID without the required scope-denial outcome	Check visible current/historical student and assignment scope before filtering; foreign existing IDs return audited 403 and absent IDs return 404	`AuthorizationIntegrationTests.test_faculty_cannot_read_export_or_modify_another_class` initially exposed three 400-vs-403 subtest failures; complete method passed after the correction
Untrusted CSV labels could begin spreadsheet formulas	Escape dangerous textual prefixes, including whitespace/control-character variants; retain numeric types	`AttendanceAlertAndReportIntegrationTests.test_csv_formula_prefixes_are_escaped_without_changing_numeric_cells` passed
Invalid/no-result exports could create misleading downloads, and export response/audit handling was incomplete	Return validation/no-results HTML without attachment; disable cache and record successful exports with format and row count	`AttendanceAlertAndReportIntegrationTests.test_exports_are_private_audited_and_invalid_or_empty_exports_are_not_downloads` passed
POSIX-only private-storage primitives did not establish Windows backup safety	Add Windows protected ACLs, native opened-handle ownership/ACL checks, reparse-point rejection and atomic non-replacing publication	All four native `apps.backups.test_storage` tests passed; encrypted backup/restore integration methods also passed
A health request carrying an authenticated cookie could load database-backed session data before readiness handled a database outage	Dispatch the public health handler at the maintenance gate before authentication/session-dependent middleware; return generic 503 for database failure	`ReadinessTests.test_readiness_never_loads_authenticated_cookie_from_an_unavailable_database` passed in the final 462-test regression run

The measured common-request and submission latency misses are tracked separately as open performance defect #49. In the initial 50-user, 1,500-request run, all common responses succeeded but only 34.6% completed within 2 seconds; p95 was 4.163 seconds. The initial marking samples were invalid because the HTTPS request harness omitted its Origin/Referer; this was a harness correction, not an application authorization defect.

The full corrected run C kept the same application source and targets, with native CSRF enforcement. All 1,800 measured requests succeeded and peak in-flight concurrency reached 50. The latency defects remain: only 22.4% of 1,500 common responses completed within 2 seconds (p95 3.702 seconds), and the three valid 60-student submissions took 5.780, 4.815 and 3.798 seconds against a 3-second limit. Each submission saved all 60 marks; first-save callbacks produced 60 notifications and 60 captured emails with no repeats on the next saves. Three semester reports passed at no more than 2.998 seconds. Both original and corrected evidence are retained in the execution record.

Monthly availability is an outstanding observation requirement rather than a corrected code defect. Performance defect #49 and the missing monthly observations keep BL-29 and release acceptance open; passing integration checks does not resolve them.